

Total No. of Questions : 10]

SEAT No. :

P3985

[5561]-689

[Total No. of Pages : 2

B.E. (Computer Engineering)
INFORMATION AND CYBER SECURITY
(2015 Pattern) (Semester - II) (410251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right side indicate full marks.*

- Q1)** a) List and explain various elements of Information security. [5]
b) Using Hill Cipher encrypt the message 'ESSENTIAL'. The key for encryption is 'ANOTHERBZ'. [5]

OR

- Q2)** a) What is steganography? What are the applications and limitations of steganography? [5]
b) Use Transposition Cipher to encrypt plain text 'I Love my India' and use the key 'HEAVEN'. [5]
[Use single columnar transposition]

- Q3)** a) Discuss elliptic curve cryptography in detail. [5]
b) What is block Cipher? Explain counter mode of block Cipher. [5]

OR

- Q4)** a) What is authentication? Explain various methods authentication. [5]
b) Explain working of AES in detail. [5]

- Q5)** a) Discuss the working of IPSec. What are the benefits of IPSec. [6]
b) What is VPN? Explain types of VPN. [6]
c) Compare PGP, MIME and S/MIME. [6]

OR

P.T.O.

- Q6)** a) Differentiate between IP-V4 and IP-V6. [4]
b) Explain secure socket layer handshake protocol in detail. [7]
c) Explain ISAKMP protocol of IPSec with header format. [7]

- Q7)** a) What are the various types of firewall. Discuss limitations of firewall. [8]
b) Explain any two password management practices. [4]
c) What is trusted system. [4]

OR

- Q8)** a) Explain need and challenges of intrusion detection system. Define signature based IDS. [8]
b) What is access control security services. [4]
c) Explain packet filtering firewall. [4]

- Q9)** a) Explain personally identifiable information PII. Describe PII impact levels with examples. [8]
b) What is cyber stalking? How to identify and detect cyber stalking. [8]

OR

- Q10)** a) What are different phases of cyber forensics? Explain with suitable diagram. [8]
b) Discuss PII confidentiality safeguards. [8]
